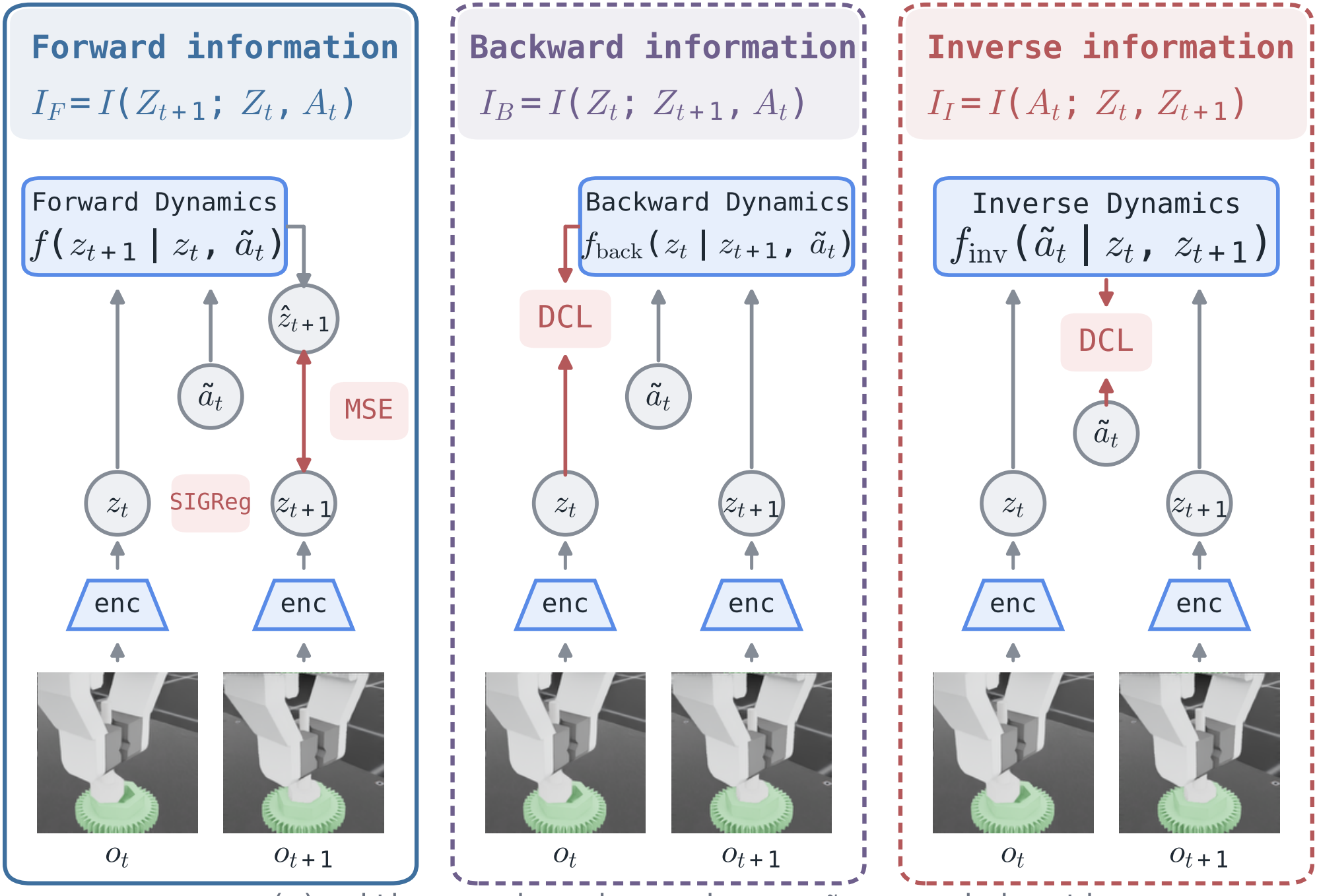

Forward information
$I_F = I(Z_{t+1};\ Z_t, A_t)$
Forward Dynamics
$f(z_{t+1} \mid z_t, \tilde{a}_t)$
$\hat{z}_{t+1}$
$\tilde{a}_t$
MSE
z_t
SIGReg
z_{t+1}
enc
enc
o_t
o_{t+1}
Backward information
$I_B = I(Z_t;\ Z_{t+1}, A_t)$
Backward Dynamics
$f_{\text{back}}(z_t \mid z_{t+1}, \tilde{a}_t)$
DCL
$\tilde{a}_t$
z_t
z_{t+1}
enc
enc
o_t
o_{t+1}
Inverse information
$I_I = I(A_t;\ Z_t, Z_{t+1})$
Inverse Dynamics
$f_{\text{inv}}(\tilde{a}_t \mid z_t, z_{t+1})$
DCL
$\tilde{a}_t$
z_t
z_{t+1}
enc
enc
o_t
o_{t+1}
$z = \text{enc}(o)$ with one shared encoder; $\tilde{a}_t$: encoded action